



PES University, Bangalore

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UE22CS251B: Microprocessor and Computer Architecture

Question Bank: Subjective Type

Sl.No	Topic	Question	Grade
1.	ARM 7TDMI: Unit-1	a. Given that LDR R1, =0x10000000 ;starting address of data transfer, R4 contains 0x10, R5 contains 0x20, R6 contains 0x30 What will be the content of register R1 and memory locations 0X10000004, 0X10000008 and 0X1000000C after the execution of the following instruction STMIA R1!, {R4-R6} b. Given that LDR R1, =0x10000000 ;starting address of data transfer, R4 contains 0x10, R5 contains 0x20, R6 contains 0x30 What will be the content of register R1 and memory locations 0X10000000, 0X10000004 and 0X10000008 after the execution of the following instruction STMIB R1!, {R4-R6}	Medium
2.	ARM 7TDMI: Unit-1	Write a program in ARM7TDMI-ISA to achieve the following. Copy blocks of 1024 bytes from source address A to a destination address B. Given that register R9 points to the start of the source data while register R10 points to the start of the destination location. Register R11 indicates the end of the source data. { Hint: Use 8 General Purpose Registers to transfer once. }	Medium
3.	ARM 7TDMI: Unit-1	Explain how stack concept is used for passing parameters in ARM. Write an ARM assembly language instruction(s) to push R4 onto the program stack referenced by SP and pop the element from SP to be copied in R5. Solution: The Memory access can also be done in LIFO order i.e Can be treated like STACK. We have a portion of memory behaving like stack. R13 is a stack pointer which will keep the address of TOP of the STACK. Mainly used in Procedural Call. Stack can grow upward or downward direction based on the MODE used by the user in the program. The Multiple data transfer instructions provide a mechanism for storing state on the runtime stack (pointed to by the stack pointer, r13 or sp). STM instruction is used to PUSH on to the STACK whereas LDM is used to POP from the STACK. Instruction to push the value in stack STMFD R13! {R4} or STMFD SP! {R4} Instruction to Pop the value from stack LDMFD R13! {R5} or LDMFD SP! {R5}	Medium

4.	ARM 7TDMI: Unit-1	Given the following code snippet in C programming language. <pre> i=1; While(i<=8) { g = h + A[i] ; i++ } </pre> g: Register R1, h: Register R2, Register R3: base address of A use LDR and ADD instructions to convert this to ARM Assembly language	Easy
5.	ARM 7TDMI: Unit-1	What are banked registers? Show how the banked registers are utilized when the user mode changes to IRQ mode.	Medium
6.	ARM 7TDMI: Unit-1	Consider the register R0 and R1 as input registers which branches to subroutine swap. Write a program for above scenario which swap 2 numbers without using SWP instruction. Let the starting address of program of 32 bit architecture be 0X00003000 . Explain how the contents of register LR and PC change when Branch instruction is executed in the program and control is returned back to the program. ANS: <pre> .text 0X00003000 Mov R0,#10 0X00003004 Mov R1, #20 0X00003008 STMFD R13!,{r0,r1} 0X0000300C BL SWAP 0X00003010 SWI 0X011 0X00003014 SWAP: 0X00003014 LDMFD R13!,{r0,R1} 0X00003018 Add r0,r0,r1 0X0000301C Sub r1,r0,r1 0X00003020 Sub r0,r0,r1 0X00003024 BX LR </pre> The address of label and LDMFD is same. When BL swap is executed: 1. PC=0X00003010 2. LR=0X00003010 PC=0X00003014 When control is returned back BX LR PC=0X00003010	

7.	ARM 7TDMI: Unit-1	What is the hexadecimal representation of the instruction “BEQ Assignment”. Indicate the address of all instructions in program. Assume the starting address of program as 0X0000103C. Given Instruction format and condition code for always=1110 and EQ=0000 31 28 27 25 24 23 0 <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%; text-align: center;">Cond</td> <td style="width: 10%; text-align: center;">101</td> <td style="width: 5%; text-align: center;">L</td> <td style="width: 75%; text-align: center;">offset</td> </tr> </table> <pre> .text mov r1, #3 BEQ assignment add r2, r0, r1 add r2,r0,r1 add r0,r0,#1 swi 0x011 Assignment: mov r0, #2 bx lr .end Answer: 0X0000103C mov r1, #3 0X00001040 BEQ assignment 0X00001044 add r2, r0, r1 0X00001048 add r2,r0,r1 0X0000104C add r0,r0,#1 0X00001050 swi 0x011 0X00001054 Assignment: 0X00001054 mov r0, #2 0X0000105C bx lr .end Cond:0000 L=0 Offset =3 000010100000.....011=0X0A000003 </pre>	Cond	101	L	offset	Medium
Cond	101	L	offset				
8.	ARM 7TDMI: Unit-1	Describe the process used for passing parameter in the ARM procedure call. Using an ALP procedure call standard, write a subroutine MAX that takes two parameters a and b and returns the larger of the two. Solution: Block transfer instructions can be used to pass parameters from set of registers or stack memory to the procedure called. <pre> .data A: .Word 10 B: .Word 20 C: .Word 0 .text LDR R0, =A LDR R1, =B LDR R2, =C LDR R3, [R0] LDR R4, [R1] </pre>	Difficult				

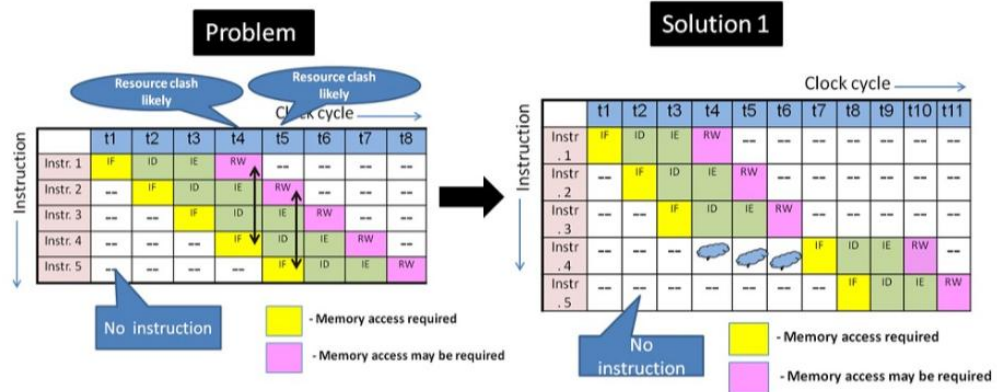
		STMFD R13!, {R3, R4} BL MAX STR R5, [R2] SWI 0X11 MAX: LDMFD R13!, {R6, R7} CMP R6, R7 BGT LOOP2 MOV R5, R7 MOV PC, LR LOOP2: MOV R5, R6 MOV PC, LR	
9.	ARM 7TDMI: Unit-1	Explain how stack concept is used for passing parameters in ARM. Write an ARM assembly language program demonstrating the use of stack in nested procedure calls. Also, use instructions LDM and STM for push and pop operations. Solution: The Memory access can also be done in LIFO order i.e Can be treated like STACK. We have a portion of memory behaving like stack. R13 is a stack pointer which will keep the address of TOP of the STACK. Mainly used in Procedural Call. Stack can grow upward or downward direction based on the MODE used by the user in the program. The Multiple data transfer instructions provide a mechanism for storing state on the runtime stack (pointed to by the stack pointer, r13 or sp). STM instruction is used to PUSH on to the STACK whereas LDM is used to POP from the STACK. <pre> .TEXT ; MAIN Procedure LDR R4, =A MOV R1, #11 MOV R2, #10 MOV R3, #02 STMFD R13!, {R1,R2,R3} BL SUM ; Call to ADD Procedure STR R0, [R4] SWI 0x11 .DATA A: .WORD 0 SUM: LDMFD R13!, { R4, R5,R6} ; ADD Procedure ADD R0, R4, R5 STMFD R13!, {R0,R6,LR} BL MUL ; Call to MUL Procedure MUL: LDMFD R13!, { R4, R5,LR} MUL R0, R4, R5 MOV PC, LR ; Return to main procedure </pre>	Medium

10.		Consider the following sequence of instructions in MIPS architecture. <pre> LDR R1, [R6,#40] BEQ R2, R3, LABEL2 ; BRANCH TAKEN ADD R1, R6, R4 LABEL2:BEQ R1,R2, LABEL1 ; BRANCH NOT TAKEN STR R2,[R4, #20] AND R1, R1, R4 </pre> a. Draw the pipeline execution diagram for this code, assuming there are no delay slots and that branches execute in the EX stage. b. Repeat the exercise mentioned in a and draw the pipeline execution diagram for this code, assuming that delay slots are used by writing a "SAFE INSTRUCTION" in the delay slot.	
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Unit-2 Pipelining

1.	Pipelining: Structural Hazard : Unit-2	Which combinations of the instructions below are involved in a Structural Hazard when handled by a 5-stage pipeline processor? a. Instruction 1: add R2, R3, R4 b. Instruction 2: ldr R5,[R6] c. Instruction 3: sub R7, R8, R9 d. Instruction 4: sub R10,R11, R1 e. Instruction 5: str R2, [R12]	Medium
2.	Pipelining: Structural Hazard : Unit-2	Observe the following figure. In any system, instruction is fetched from memory in IF machine cycle. In the 4-stage pipeline shown Result Writing (RW) may access memory or one of the General-Purpose Registers depending on the instruction. At t4, Instruction-1(I1) is at RW stage and Instruction-4(I4) at IF stage. Alas! Both I1 and I4 are accessing the same resource i.e memory if I1 is a STORE instruction. Explain if it is possible to access memory by 2 instructions from the same CPU in a timing state? What dependency does it indicate? What is the solution? Ans:	Medium

Solution 1: Introduce bubble which stalls the pipeline as in figure below



3.	Pipelining: Structural Hazard : Unit-2	Consider an instruction pipeline for the MIPS32 processor where data references constitute 42% of the instructions, and the ideal CPI ignoring memory structural hazards is 1.25. How much faster is the ideal machine without the memory structural hazard versus the machine with the hazard?	Me diu m
4.	Pipelining: Unit-2	Loads/stores (MEM) use same memory port as instruction fetches (IF). Given that 30% of all instructions are loads and stores. Assume CPI old is 1.5 1 2 3 4 5 6 7 8 9 i IF ID EX MEM WB <— a load i+1 IF ID EX MEM WB i+2 IF ID EX MEM WB i+3 ** IF ID EX MEM WB i+4 IF ID EX MEM WB How much faster could a new machine with two memory ports be?	

8.	Pipelining Hazards : Unit-2	What are the different types of hazards observed during pipelining? Explain how to overcome the same. Answer: • Structural Hazard: • Due to structure of the pipeline • Hardware cannot support combination of instructions in the pipeline needing the same resources Solution: Additional Hardware is required, Harvard Architecture for memory. • Data Hazard: • Due to data dependencies between instructions • Instruction depends on the result of prior instruction still in the pipeline In Software Solution 1: Re-order instructions Solution 2: insert independent instructions (or no-ops) Ex: MOV R0, R0 In Hardware Solution 1: Insert bubbles (i.e. stall the pipeline) Solution 2: Data Forwarding • Control Hazard: • Due to control instructions • Pipelining of branches & other instructions that change the PC Safe Instruction, Delayed Branching, Bringing the execution of branch instruction earlier in the pipelining, Branch Prediction.	Medium
9.	Pipelining Hazards : Performance : Unit-2	Given that two programs P1 and P2 have an ideal CPI value equal to 1. For P1, 10% of the instructions have a load-use hazard and 15% of its instructions are taken branches. For P2, 20% of the instructions have a load-use hazard, and 5% of its instructions are taken branches. (Note: For a load use hazard one stall cycle is needed, and for a taken branch two stall cycles are needed) What is the performance comparison between P1 and P2? a) CPI of P2 is less than the CPI of P1, Hence P2 is faster b) CPI of P1 is less than the CPI of P2, Hence P1 is faster c) CPI of P2 is equal than the CPI of P1, hence P1 and P2 are equal. d) CPI of P2 is greater than the CPI of P1, Hence P1 is faster Explanation $CPI\ of\ P1 = 1 + 0.1 * 1 + 0.15 * 2 = 1.4$ $CPI\ of\ P2 = 1 + 0.2 * 1 + 0.05 * 2 = 1.3$ The CPI of P2 is less than the CPI of P1. Hence, P2 is faster	
10.	Pipelining Hazards : Branch Prediction:	Explain the Branch History Table that is inevitable to predict in 1st stage of pipeline processor.	

	Unit-2		
10		We want to perform the operation $Y * 253$, Register R1 holds the value of Y and Register R2 holds the value 253. We expect register R1 to hold the result of the operation. Write ARM assembly program to perform the operation, without using the MUL or MLA instruction. Try to use as few instructions as possible.	
11.		;IF(R0==R1)R2++ ELSE R3--; USE CONDITIONAL EXECUTION and choose the correct code snippet in ARM-ISA. Answer: a. MOV R0,#5 MOV R1,#5 CMP R0,R1 ADDEQ R2,R2,#1 SUBNE R3,R3,#1 SWI 0X011	
12		Write a program to convert the following HLL to assembly language, assume R1 = 0x9, R2 = 0x6 and R3 = 0x5 If R1>R2 Then R3=R3-5	
13		Write a program in ARM7TDMI-ISA to check the parity of given 32 bit number using function subprogram PARITYCHECK. Display appropriate messages as ODD PARITY or EVEN PARITY number. .DATA ODD: .ASCIZ "ODD PARITY NUMBER" EVEN: .ASCIZ "EVEN PARITY NUMBER" A: .WORD 0X12345678 .TEXT LDR R2,=A LDR R3,[R2] MOV R6,#0 ; COUNT REGISTER MOV R4,#0 ; COUNT NUMBER OF 1S. L2: TST R3,#1 BNE L1 LOOP:MOV R3,R3,LSR #1 ADD R6,R6,#1 CMP R6,#32 BNE L2 TST R4,#1 BEQ L3 LDR R0,=ODD SWI 0X02 B EXIT L3: LDR R0,=EVEN SWI 0X02	

		EXIT:SWI 0X011 L1:ADD R4,R4,#1 B LOOP .END	
14		Let register R0=0xFFFFFFFF and R13=0x008010. Show the content of stack and update on stack pointer when following instructions are executed. i.STR R0,[SP,#-8] ii. STR R0,[SP,#-8]!	

Additional questions:

A five-stage pipeline has stage delays of 150, 120, 150, 160 and 140 nanoseconds. The registers that are used between the pipeline stages have a delay of 5 nanoseconds each.

The total time to execute 100 independent instructions on this pipeline, assuming there are no pipeline stalls, is _____ nanoseconds.

Data:

Number of Instructions = $n = 100$

Number of stages = 5;

Stage delay = 5 ns

Calculation:

Time taken by five stage pipeline processor of singel instruction = $T = \text{Max}(150, 120, 150, 160, 140) + \text{stages delay}$
 $= 160 + 5 = 165 \text{ ns}$

The time required to execute n instructions with pipeline = $[k + (n - 1)]T$
 $= (5 + (100 - 1)) \times 165 = 17160 \text{ ns}$

The speed gained by an 'n' segment pipeline executing 'm' tasks is:

1. $\frac{(n+m-1)}{mn}$

2. $\frac{mn}{(n+m-1)}$

3. $\frac{n+m}{(mn-1)}$

4. $\frac{n+m}{(mn+1)}$

Ans: option 2

Consider a 6-stage instruction pipeline, where all stages are perfectly balanced. Assume that there is no cycle-time overhead of pipelining. When an application is executing on this 6-stage pipeline, the speedup achieved with respect to non-pipelined execution if 25% of the instructions incur 2 pipeline stall cycles is _____.

Instruction pipeline = 6 stage

25% of the instructions incur 2 pipeline stall cycles

Formula:

$$\text{Speed up} = \frac{\text{Execution time}(\text{non pipeline})}{\text{Execution time}(\text{pipeline})}$$

Time with pipeline = $1 + \text{stall frequency} \times \text{stall cycle}$

Calculation:

Time without pipeline = 6 stages = 6 cycles

Time with pipeline = $1 + .25 \times 2 = 1.5$

$$\text{Speed up} = \frac{6}{1.5} = 4$$

Speedup = pipeline depth / $1 + \text{stalls}$

Consider an unpipelined machine with 10nsec clock cycles which uses four cycles for ALU operations and branches where as five cycles for memory operation. Assume that the relative frequencies of these operations are: 40%, 20% and 40%, respectively. Due to clock skew and setup pipeline let us consider that the machine adds one nsec overhead to the clock. How much speedup is observed in the instruction execution rate when a pipelined machine is considered.

EXPLANATION:

Operation	Frequency	Clock
ALU	40% = 0.40	4
BRANCH	20% = 0.20	4
MEMORY	40% = 0.40	5

Total Clock Cycle = $0.40 * 4 + 0.20 * 4 + 0.40 * 5$

Total Clock Cycle = $1.6 + 0.8 + 2$

Total Clock Cycle = 4.4 Clock Cycle

Here,

Non-pipeline time = 4.4 clock cycle * 10 nsec

Non-pipeline time = 44nsec

Pipeline time = Clock time + overhead time = 10nsec + 1nsec = 11nsec

Speedup= Time taken by non-pipeline machine / Time taken by pipeline machine

Speedup = $44/11 = 4$

So, the correct answer is 4 times

A non-pipeline system takes 50ns to process a task. The same task can be processed in six-segment pipeline with a clockcycle of 10ns. Determine approximately the speedup ratio of the pipeline for 500 tasks.

1. 6
2. 4.95
3. 5.7
4. 5.5

Speed up factor is defined as the ratio of time required for non-pipelined execution to that of time received for pipelined execution.

Data:

Time for non-pipelined execution per task = $t_n = 50$ ns

Time for pipelined execution per task = $t_p = 10$ ns

Number of stages in the pipeline = $k = 6$

Number of tasks = 500

Formula

$$S = \frac{T_n}{T_p}$$

S = speed up factor

Calculation:

Time for non-pipelined = $T_n = t_n \times \text{Number of tasks}$

Time for non-pipelined = $T_n = 50 \times 500$

Time for pipelined = $T_p = 1\text{st task} \times k \times t_p + (\text{All Remaining Tasks } (k - 1)) \times t_p$

Time for pipelined = $T_p = 1 \times 6 \times 10 + (500 - 1) \times 10$

$$S = \frac{T_n}{T_p} = 4.95$$

1. Consider the following instruction set:

LDR r3,[r4]

ORR r5,r6,r7

Statement 1: ORR does nothing in MEM Stage.

Statement 2: Skipping MEM Stage does not cause structural Hazard.

- a. Both statement is true
- b. Both statement are false
- c. Statement 1 is True and statement 2 is false
- d. Statement 1 is False and statement 2 is True

Ans: c

2. If there is unified memory in pipeline then every Overlap of load/store with Instruction Fetch will introduce how many stalls?

- a. 4
- b. 5
- c. 6
- d. 3

Ans: 3

3. Which of following statements are True?

Statement 1: PC will be copied to R14 the Link Register (LR) after branch is taken

Statement 2: The functionality of BX LR is same as MOV PC,LR

- a. Both statements are true
- b. Both statements are false
- c. Statement 1 is false and statement 2 is true
- d. Statement 1 is True and statement 2 is False

Ans: C

4. Suppose address of x is indicated by register R0 and address of y is indicated by register R1. What is the ARM equivalent to the statement $x=y$

a) LDR R0,[R1]

STR R1,[R0]

b) LDR R1,[R1]

STR R0,[R0]

c) LDR R2,[R1]

STR R2,[R0]

d) LDR R0,[R0]

STR R1,[R1]

5. _____ symbol is used to signify write back mode.

a) #

b) ^

c) &

d) !

6. The instruction, MLA R0,R1,R2,R3 performs _____

a) $R0 \leftarrow [R1] + [R2] + [R3]$

b) $R3 \leftarrow [R0] + [R1] + [R2]$

c) $R0 \leftarrow [R1] * [R2] + [R3]$

d) $R3 \leftarrow [R0] * [R1] + [R2]$

7. What is the output after execution of the following instruction

LDMIA r10! {r0, r1, r4, r10}

- a) Content of memory location pointed by r10 is copied to r0,
Content of memory location pointed by $r10 + 4$ is copied to r1
Content of memory location pointed by $r10 + 8$ is copied to r4

Content of memory location pointed by $r10 + 12$ is copied to $r10$

b) This is an invalid instruction

c) Content of Register $R0$ is copied to memory location pointed by $r10$

Content of Register $R1$ is copied to memory location pointed by $r10 + 4$

Content of Register $R4$ is copied to memory location pointed by $r10 + 8$

Content of Register $R10$ copied to memory location pointed by $r10 + 12$

d) The content of consecutive locations in stack is saved on to $r0, r1, r4, r10$

8. In instruction pipelining, main memory has a single port and that all instruction fetches and data reads must be performed at one time (simultaneously). In this case, an operand read to or write from memory cannot be performed in parallel with an instruction fetch and an ignore cache. Which of the following hazard represents the above situation?

a) Resource Hazard

b) Data Hazard

c) Control Hazard

d) R-W Hazard

9. In instruction pipelining, main memory has a single port and that all instruction fetches and data reads must be performed one at a time. In this case, an operand read to or write from memory cannot be performed in parallel with an instruction fetch and an ignore cache. Which of the following hazard represents the above situation?

a) Resource Hazard

b) Data Hazard

c) Control Hazard

d) R-W Hazard

10. The contention for the usage of a hardware device is called _____

a) Structural hazard

b) Stalk

c) Deadlock

d) None of the mentioned

11. The periods of time when the unit is idle is called as _____

a) Stalls

b) Bubbles

c) Hazards

d) Both Stalls and Bubbles

12. _____ Are Used to stop Specific interrupt.

A. Interrupt mask

- B. Interrupt request
- C. Both
- D. None

ANS- A

13.CPSR has ___ interrupt mask bits.

- A. 2**
- B. 25
- C. 6
- D. 4

ANS- A

14.___ interrupt levels available on the ARM processor

A.2

B.5

C.6

D.4

ANS- A

15.___ is provided to service general-purpose interrupt handling of internal and external events.

- A. IRQ**
- B. FIQ
- C. SWI
- D. None

ANS- A

16.Which of the following is characteristic of a RISC processor?

- a. Variable length instruction formats
- b. Large variety of addressing modes
- c. Single cycle instruction execution**
- d. Large number of instructions

17.Which processors includes multi-clocks?

- a. CISC**
- b. RISC
- c. ISA
- d. None

18. A processor performing fetch or decoding of different instruction during the execution of another instruction is called _____

- a. Super-scaling
- b. Pipelining**
- c. Parallel Computation
- d. None of the mentioned

19. Pipeline overhead arises from the combination of pipeline register delay and

- a. Hit rate
- b. Clock cycle
- c. Frequency
- d. Clock skew**

20. Equivalent instruction for Full Descending stack :

- LDMDA and STMIB
- LDMIA and STMDB**
- LDMDB and STMIA
- LDMIB and STMDA

21. Consider the set of instruction:

```
.text
MOV R0,#4
MOV R1,#5
STMEA R13!, {R0,R1}
LDMEA R13!,{R5,R6}
.end
```

The value of stack pointer (assume initial SP is 5400) after the execution :

- 5400**
- 5404
- 5408
- 540C

22. The unit that preserves the history of each conditional branch is

- a) Branch Target Buffer (BTB)
- b) Branch History Table (BHT)**
- c) Symbol Table
- d) Reorder Table

23. . For the 1-bit prediction scheme, which of the following statements are false?

There will be two mispredictions whenever the behavior of a branch instruction changes.

An entry in the BPB uniquely identifies a branch instruction.

Does not give any advantage for the pipeline.

None of the above.

24. Consider a program with two branch statement with the following behavior

T T N T T N T T T T N T T T T T T N T T T T T N T

How many Miss Prediction if initial state is non taken?

14

15

13

16